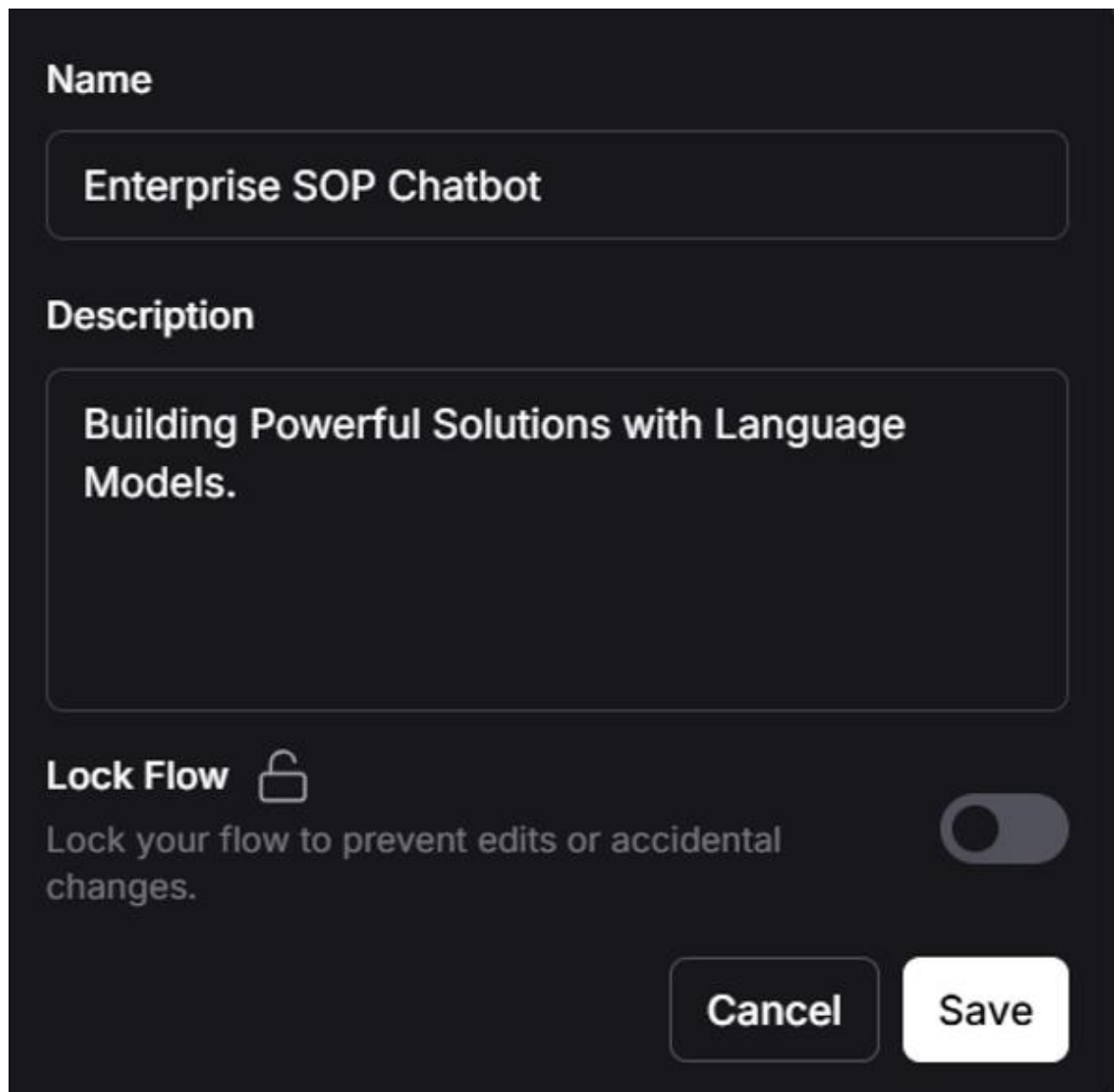


Enterprise SOP Chatbot using Langflow & Gemini

This project demonstrates the implementation of an Enterprise SOP Chatbot using Langflow, Google Gemini, Chroma Local, and Retrieval-Augmented Generation (RAG). The chatbot retrieves relevant information from uploaded SOP documents and generates grounded responses.

Step 1 – Create a New Langflow Project

Create a new Langflow project named 'Enterprise SOP Chatbot'.

A dark-themed dialog box for creating a new Langflow project. It has a 'Name' field with the text 'Enterprise SOP Chatbot', a 'Description' field with the text 'Building Powerful Solutions with Language Models.', a 'Lock Flow' section with an unlocked padlock icon and a toggle switch, and 'Cancel' and 'Save' buttons at the bottom right.

Name

Enterprise SOP Chatbot

Description

Building Powerful Solutions with Language Models.

Lock Flow 

Lock your flow to prevent edits or accidental changes. 

Cancel Save

Step 2 – Configure Google Gemini

Generate a Google Gemini API key and configure it in Langflow. Select Gemini 3.5 Flash as the language model.

API key details

 For details on using auth keys with Gemini API, see our [API key documentation](#).

API Key	
Name	
Gemini API Key	
Project name	
projects/169014560558	
Project number	
169014560558	

Delete key

Copy cURL quickstart

Copy key

Connect the Gemini API key inside Langflow.

Model providers

Q Google Generative AI

Google Gen...

28 models

Google API Key *

Configure your [Google Generative AI credentials](#) to enable these models

Alza.....

Disconnect

Replace

Q Search models...

Language Models

gemini-omni-flash-preview

reasoning

vision

gemini-3.5-flash

tool

reasoning

vision

gemini-flash-latest

tool

reasoning

vision

gemini-3.1-flash-lite

tool

reasoning

vision

gemini-flash-lite-latest

tool

reasoning

vision

gemma-4-26b-a4b-it

tool

reasoning

vision

gemma-4-31b-it

tool

reasoning

vision

Step 3 – Create the Knowledge Base

Create a Knowledge Base named SOP_DB, select Chroma Local as the vector database, and configure the embedding model.

Create new Knowledge Base

Create new knowledge in Langflow.

Knowledge Name *

Type something...

Choose Embedding Model *

↶ Connect other models

DB Provider *

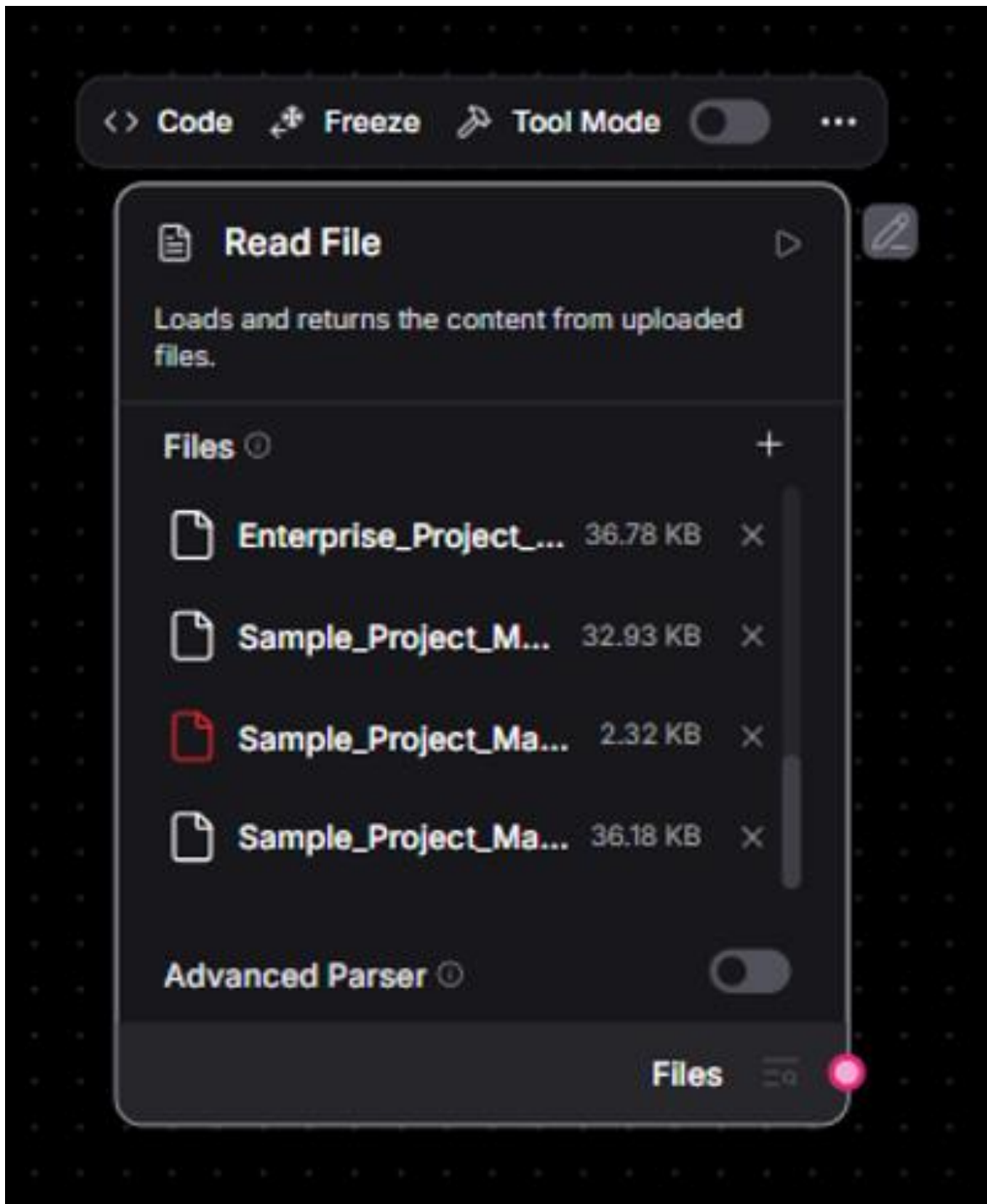
Chroma Local

Cancel

Create

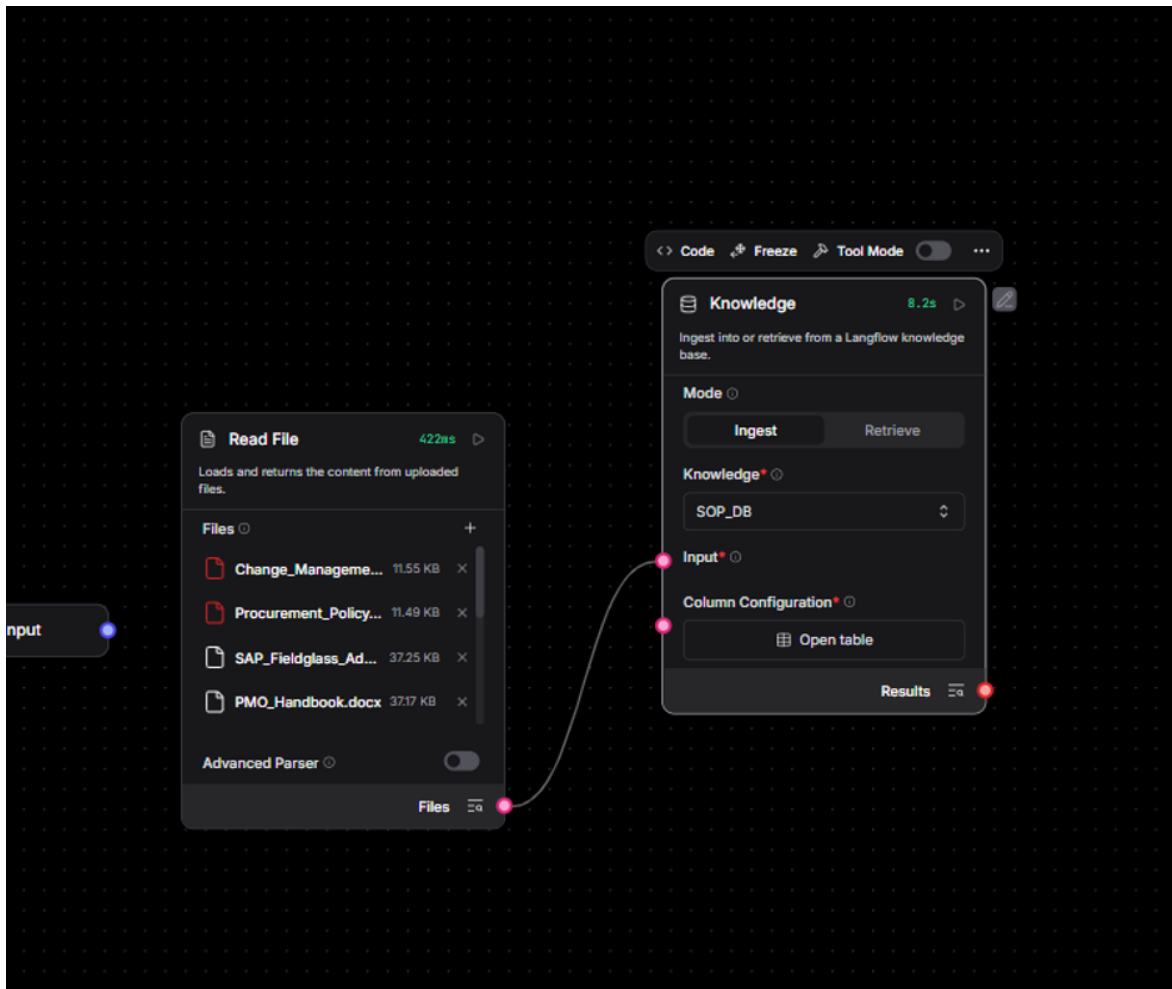
Step 4 – Upload SOP Documents

Upload the enterprise SOP documents (PDF/DOCX) using the Read File component.



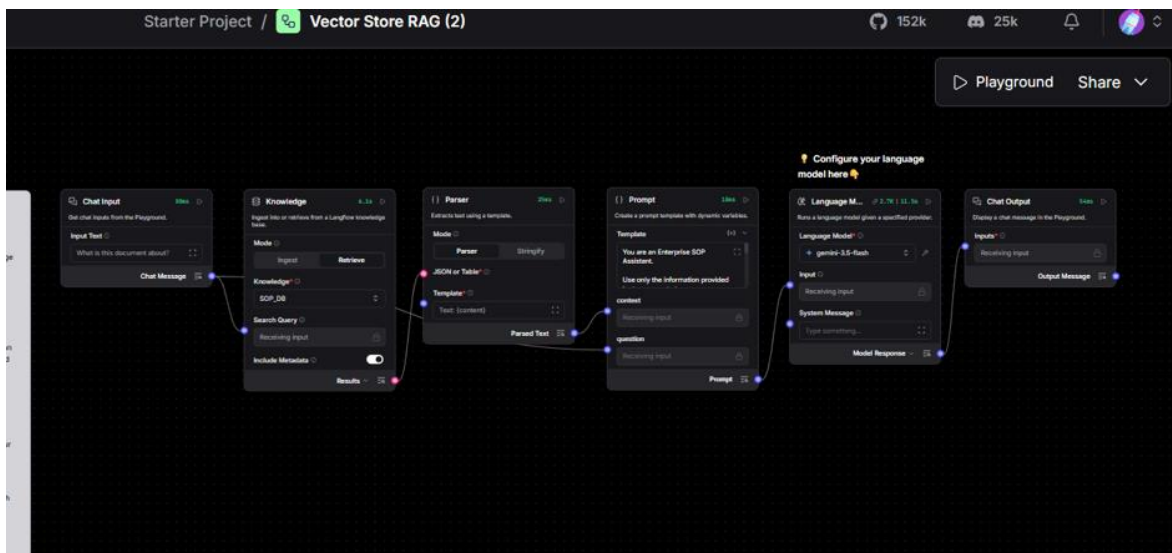
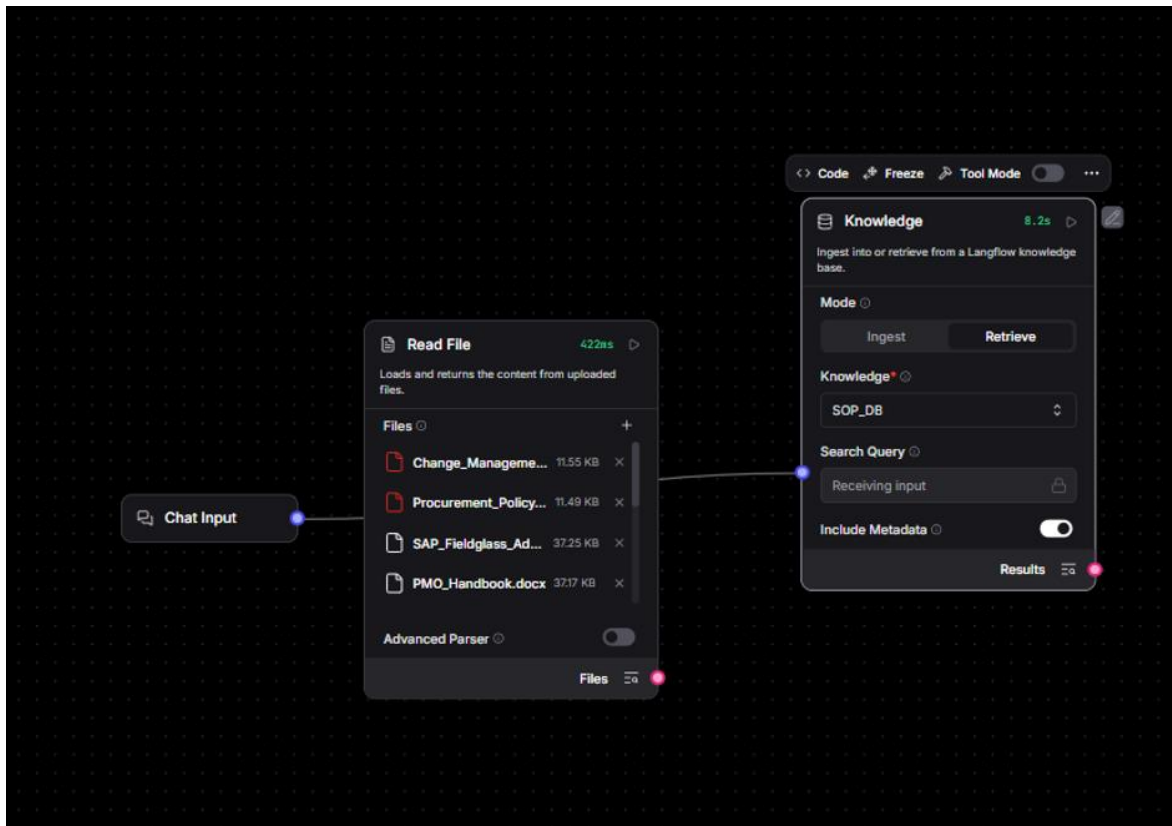
Step 5 – Ingest Documents

Connect Read File to Knowledge (Ingest). Langflow extracts the document text, creates embeddings, and stores them in the SOP_DB vector database.



Step 6 – Configure the Retrieval Flow

Use the Vector Store RAG template to build the chatbot workflow. This workflow retrieves relevant information from the knowledge base before sending it to Gemini to generate an accurate response.



- **Chat Input:** Receives the user's question.
- **Knowledge (Retrieve):** Searches SOP_DB and retrieves the most relevant document chunks using semantic search.
- **Parser:** Formats the retrieved content into a structured format.
- **Prompt:** Combines the user's question, retrieved document context, and system instructions into a single prompt.

- **Gemini 3.5 Flash:** Generates an answer using the retrieved SOP content, reducing hallucinations.
- **Chat Output:** Displays the final response to the user.

Workflow:

User Question



Chat Input



Knowledge (Retrieve)



Parser



Prompt



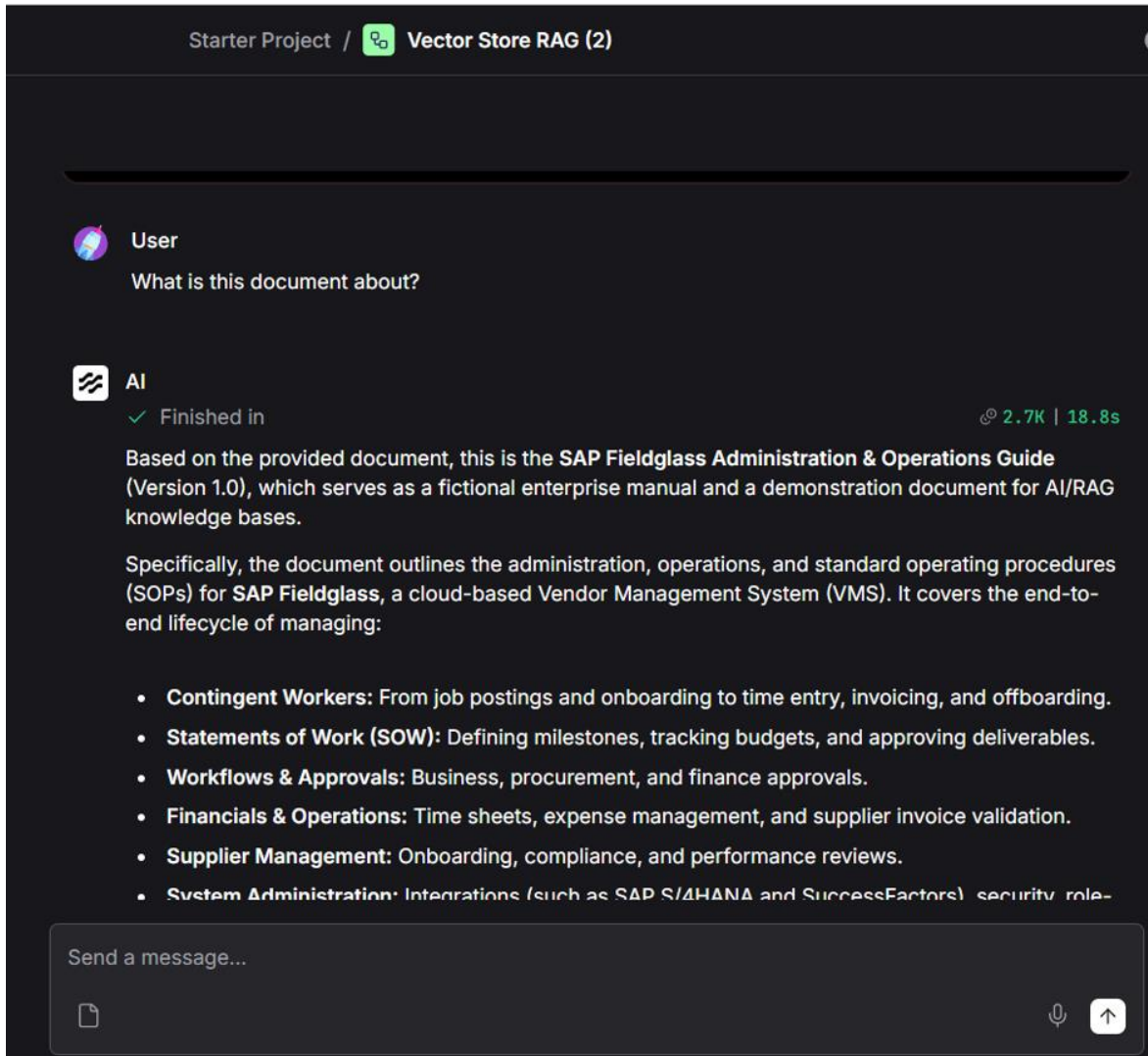
Gemini 3.5 Flash



Chat Output

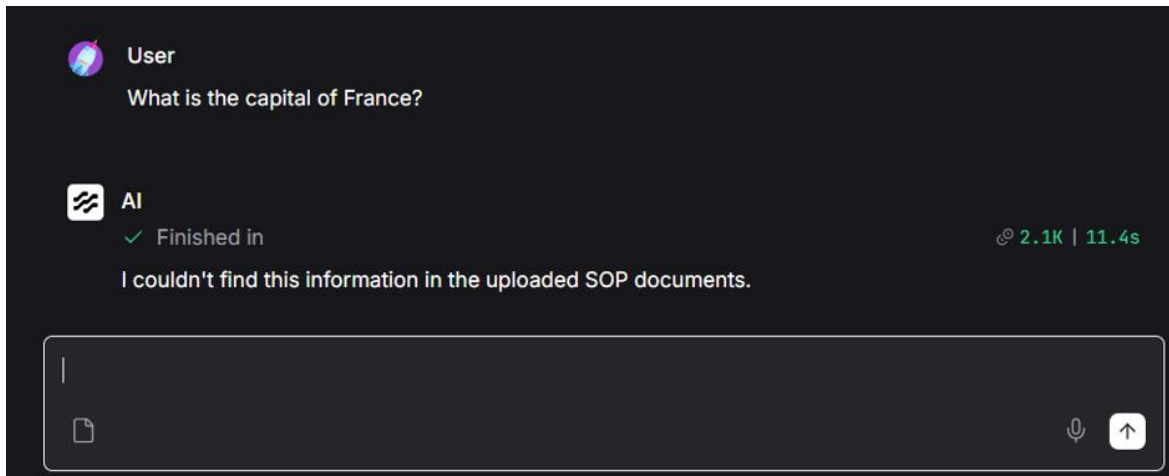
Step 7 – Test the Chatbot

Ask a question related to the uploaded SOP documents. The chatbot retrieves the relevant content and generates an accurate response.



Step 8 – Validate the RAG System

Ask a question that is not covered in the uploaded documents. The chatbot correctly states that the information is unavailable, confirming that responses are grounded in the knowledge base.



Technology Stack

- Langflow
- Google Gemini 3.5 Flash
- Chroma Local
- RAG
- PDF/DOCX Documents

Future Enhancements

Integrate SharePoint, OneDrive, or Amazon S3 for automatic document ingestion, schedule document indexing, and deploy the solution for enterprise use.